

Eissa Islam

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years building production-grade systems across LLMs, RAG pipelines, computer vision, and MLOps. Specialize in end-to-end delivery — from architecture design to cloud deployment — on real projects at a university, fintech, and government initiatives. Published IEEE author (ITC-Egypt 2024). Proven track record in agentic AI workflows, autonomous agents, conversational AI, and data-intensive applications that solve tangible problems.

EXPERIENCE

AI Engineer — Misr University for Science & Technology · Giza, Egypt Feb 2025 – Present

- Spearheading university-wide smart infrastructure initiatives, delivering 3 production AI systems currently in active campus use.
- Built RoboMUST, a voice-enabled AI learning platform supporting students and professors via RAG, local LLMs, quiz generation, and smart summarization.
- Developed ShopIntel, an LLM-powered shopping assistant with real-time multi-platform web scraping, price comparison, and personalized planning via NLP pipelines.
- Deployed a campus-wide facial recognition attendance system covering multiple departments, fully automating manual record-keeping workflows.
- Built a YOLOv8-based garbage classification system for smart campus recycling deployed at physical sorting stations.

AI Engineer — AiTech · Egypt Oct 2024 – Dec 2024

- Architected an intelligent chat system using web scraping, MongoDB, ChromaDB, and RAG for context-aware LLM-driven customer interactions.
- Built a Quranic RAG chatbot grounded in classical tafsir sources using a dual-vector pipeline (MongoDB + ChromaDB) for semantically accurate verse retrieval.
- Developed ScrapeMaster, a Tkinter GUI desktop tool for automated YouTube metadata and web image scraping used for internal dataset collection.
- Collaborated cross-functionally to ensure scalability and reliability in production under tight delivery timelines.

ML Engineer Intern — Digital Egypt Initiative (MCIT) · Egypt Apr 2024 – Oct 2024

- Gained hands-on MLOps experience with MLflow, Hugging Face, and Azure AI on 4 applied projects including fraud detection and NLP pipelines.
- Built end-to-end data pipeline and model deployment workflows resulting in deployment-ready, evaluated ML models.

Data Scientist Trainee — Banque Misr — Rowad Internship · Egypt Jun 2024 – Aug 2024

- Designed fraud detection and credit scoring models on imbalanced datasets using SMOTE and ensemble methods to maximize recall without sacrificing precision.
- Delivered feature engineering reports and collaborated with analysts and data engineers in an Agile environment.

Data Science Trainee — ALX Africa & ExploreAI · Remote May 2023 – Aug 2024

- Completed capstone projects involving model training, evaluation, and dashboard development using deep learning and data science techniques on real-world datasets.
- Applied supervised and unsupervised learning methods across structured and unstructured data problems.

ML Intern — Misr University AI Tech R&D Center · Giza, Egypt Jan 2023 – Jul 2023

- Developed a deep learning prostate cancer detection model in TensorFlow/Keras, leading all phases from DICOM preprocessing to final performance validation.

EDUCATION

B.Sc. Information Technology — Major: Artificial Intelligence

Misr University for Science and Technology, Giza, Egypt | GPA: 3.41 / 4.0

Sept 2019 – June 2024

TECHNICAL SKILLS

Languages & DBs	Python, SQL, MongoDB, ChromaDB, PostgreSQL, SQLite
ML / DL	TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face Transformers
LLM & RAG	LangChain, LangGraph, Ollama, Gemini API, Groq (Llama 3), BGE-m3, Semantic Search, Re-ranking, Prompt Engineering
MLOps & Cloud	MLflow, Azure AI, AWS, Oracle Cloud (ARM), GitHub Actions, Sentry, systemd
Web & APIs	FastAPI, Flask, Streamlit, REST APIs, SSE, Uvicorn, HTML/CSS/JS
Scraping & Auto	Scrapling, Selenium, Playwright, BeautifulSoup, httpx, n8n, APScheduler, Docling, Celery, Redis
Data & Viz	Pandas, NumPy, Matplotlib, Plotly, Power BI, Excel
Vector DBs	ChromaDB, Qdrant, Milvus, PGVector
Computer Vision	YOLOv8, OpenCV, Mediapipe, Object Detection, Pose Estimation
Soft Skills	Agile / Scrum, Technical Leadership, Research, Problem-solving

KEY PROJECTS

- JobPilot AI — Autonomous Job Application Agent** *Tech: Python 3.11, Scrapling, Playwright, Gemini API, Groq/Llama 3, SQLite, SQLAlchemy, Alembic, Streamlit, python-telegram-bot, Jinja2, python-docx, FPDF2, APScheduler, Docling, Oracle Cloud, GitHub Actions, Sentry*
 - Built a fully autonomous 7-layer Python system that scrapes job postings 24/7 across Wuzzuf, Indeed, RemoteOK, and WeWorkRemotely using async anti-bot stealth scraping with fuzzy cross-platform deduplication.
 - Engineered a hybrid scoring engine combining rule-based filtering (keywords, location, seniority, salary) with LLM semantic analysis (Gemini API + Groq/Llama 3 fallback) to score each job 0–100 against a candidate profile.
 - Built an AI-powered CV tailoring module using prompt-engineered LLM calls to generate ATS-optimized DOCX and PDF files via Jinja2 templates — producing 80+ tailored CVs without fabricating skills.
 - Implemented a Telegram bot for real-time notifications, daily digests, and human-in-the-loop approval gates before any application is submitted.
 - Deployed a multi-page Streamlit dashboard with job board, application tracker, Skills Gap Analyzer, and automated PDF weekly reports — hosted on Oracle Cloud Always-Free ARM VM at zero cost via systemd + GitHub Actions CI/CD.
- Enterprise Brain — AI-Powered Enterprise Knowledge & Operations System** *Tech: Python, LangGraph, FastAPI, BGE-m3, Qdrant, Celery, Redis, Ollama, Docling, Webhooks, Google Drive API, Slack API, React/Next.js, SSE*
 - Architected a multi-agent RAG system that transforms enterprise data sources (Google Drive, Slack, Notion, Jira) into a queryable knowledge base — enabling employees to retrieve information and trigger actions through a single conversational interface.
 - Built a multi-modal ingestion pipeline handling text documents, Excel tables, scanned images (OCR), and meeting audio files, with real-time Webhook-driven processing via Celery + Redis background workers.
 - Engineered a LangGraph-based agentic orchestration layer with a Router/Supervisor Agent, a Retrieval Agent using BGE-m3 embeddings and semantic re-ranking for high-precision context, and an Action Agent capable of executing multi-step tasks (e.g., summarize report → compare data → send email).
 - Implemented enterprise-grade RBAC security ensuring users can only query vectors their role permits, preventing unauthorized access to sensitive HR, payroll, or executive data.
 - Delivered a React dashboard with real-time SSE streaming, clickable source citations, and human-in-the-loop approval gates for any AI-proposed actions before execution.

RoboMUST — AI-Powered Educational Platform *Tech: Python, FastAPI, HTML/CSS/JS, MongoDB, ChromaDB, LangChain, Ollama, Whisper, TTS, RAG, Docling*

- Built an end-to-end educational AI platform enabling students and professors to interact with course material via text or voice — live in production at Misr University.
- Implemented RAG pipelines over course PDFs with local LLMs supporting QA, quiz generation, and smart summarization.
- Delivered full-stack solo: AI logic, FastAPI backend, MongoDB/ChromaDB storage, REST APIs, and complete web interface.

ShopIntel — AI-Powered Conversational Shopping Platform *Tech: Python, FastAPI, BeautifulSoup, Selenium, MongoDB, LLMs, NLP*

- Developed AI and backend for an LLM-powered assistant enabling product search, price comparison, and multi-item studio/outfit planning across live e-commerce sites via real-time scraping pipelines.

Graduation Project: Parkinsonian Gait Severity Assessment (IEEE Published) *Tech: Python, Mediapipe, Scikit-learn, Keras, Pandas, NumPy*

- Led development of a vision-based gait analysis system for dementia patients achieving 92% validation accuracy using 2D/3D video data and a multi-model ML ensemble.
- Published at IEEE ITC-Egypt 2024 — IEEE Xplore

Quranic RAG Chatbot *Tech: Python, ChromaDB, MongoDB, LangChain, Hugging Face, Ollama*

- Built a grounded RAG chatbot answering questions about the Holy Quran using classical tafsir sources, with a dual-vector semantic search pipeline for context-rich verse retrieval.

Smart Face Attendance System *Tech: Python, dlib, Face Recognition Libraries*

- Contributed to a campus-wide facial recognition attendance system for real-time student monitoring, automating logging and record-keeping across multiple university departments.

PUBLICATION

Parkinsonian Gait Severity in Older Adults with Dementia Using Natural Gait Video Analysis

IEEE 2024 — International Telecommunications Conference (ITC-Egypt) [View on IEEE Xplore](#)